

EXPLICIT GRAM WITNESSES FOR OFF-CRITICAL PAIR BLOCKS IN FINITE WEIL QUADRATIC FORMS

MARCUS HERMANSSON

ABSTRACT. We isolate a constructive finite-dimensional certificate for the negativity generated by a transverse off-critical zero pair in a finite probe representation of Weil’s quadratic form. If the complex response of a reflection-symmetric pair decomposes as $u = x + iy$, the associated real pair block has the form

$$B = c(xx^\top - yy^\top), \quad c > 0.$$

We construct the explicit vector

$$w = \|x\|^2 y - \langle x, y \rangle x$$

and prove

$$w^\top B w = -c(\|x\|^2 \|y\|^2 - \langle x, y \rangle^2).$$

Hence every transverse pair possesses an explicit finite negative witness. The same witness separates the pair block from the cone generated by all nonnegative real rank-one blocks, the form taken by critical-line contributions in the finite model. We derive exact inertia, a normalized negative margin, operator-norm perturbation bounds, and a quantitative masking criterion, and we compare the construction with Bombieri’s finite negative-inertia theorem. The contribution is a constructive local refinement: it does not provide a new proof that off-critical zeros generate negativity, and it does not prove the Riemann hypothesis. The core witness identity, strict-negativity statement, on-line nonnegativity theorem, and representation obstruction are formally verified in Lean 4.

1. INTRODUCTION

Weil’s positivity criterion recasts the Riemann hypothesis as a positivity statement for a quadratic functional obtained from the explicit formula [8, 9, 1, 2]. A broad line of work, from Yoshida’s localized forms and Bombieri’s variational analysis to recent spectral and finite-dimensional constructions, asks what finite or localized shadows of that positivity criterion retain exact arithmetic meaning [3, 4, 5, 7].

The classical negative-index phenomenon is not new here. Bombieri proved, in a finite spectral model with the symmetries of the nontrivial zeta zeros, that the number of negative eigenvalues is exactly the number of distinct complex-conjugate pairs in the corresponding transformed zero set; under a finiteness hypothesis on off-critical zeros, this yields the eventual finite negative-inertia count for zeta truncations [1]. In particular, the implication

$$\text{off-critical spectral pair} \implies \text{finite negative inertia}$$

is established territory.

The question considered here is more local and constructive:

Date: 25 August 2026.

2020 Mathematics Subject Classification. Primary 11M26; Secondary 15A18, 47A58, 03B35.

Key words and phrases. Riemann hypothesis, Weil quadratic form, Gram determinant, positive semidefinite cone, finite truncation, off-critical zeros, formal verification.

Given the finite block contributed by one visible off-critical pair, can its negativity be certified by a closed-form vector, with an exact margin, and can that same vector separate the block from the entire cone generated by critical-line rank-one contributions?

The answer is elementary once the correct pair block is isolated. Its value is constructive: the certificate makes one negative direction explicit, quantifies its margin, and exposes the precise scalar inequality that governs masking along that direction. Let V be a finite-dimensional real inner-product space and let $x, y \in V$. For $c > 0$, consider

$$(1.1) \quad B = c(x \otimes x - y \otimes y),$$

where $(x \otimes x)z = \langle x, z \rangle x$. Define

$$(1.2) \quad \Delta(x, y) = \|x\|^2 \|y\|^2 - \langle x, y \rangle^2$$

and

$$(1.3) \quad w = \|x\|^2 y - \langle x, y \rangle x.$$

Then

$$(1.4) \quad \langle Bw, w \rangle = -c \Delta(x, y)^2.$$

Thus, whenever x and y are linearly independent, strict Cauchy–Schwarz gives $\Delta(x, y) > 0$ and w is an explicit negative witness.

The same vector yields a convex-geometric separation statement. Every finite nonnegative combination

$$P = \sum_j c_j (v_j \otimes v_j), \quad c_j \geq 0,$$

is positive semidefinite, and therefore $\langle Pw, w \rangle \geq 0$. Consequently the functional $A \mapsto \langle Aw, w \rangle$ separates B from the whole positive-semidefinite cone.

Main contributions. The paper packages this observation into the following finite theorem suite.

- (i) An explicit orthogonal witness for a signed rank-one pair block.
- (ii) The exact Gram-determinant value $-c\Delta(x, y)^2$ of the quadratic form at that witness.
- (iii) The exact inertia of an isolated transverse pair block.
- (iv) An explicit separating functional between the pair block and the positive-semidefinite rank-one cone.
- (v) A normalized negative margin and an operator-norm robustness bound.
- (vi) An exact masking criterion for a positive background added to the pair block.
- (vii) A specialization to the finite zero-pair geometry occurring in a Weil probe-response model.
- (viii) A correspondence with a Lean 4 formalization of the core witness and separation theorems.

The finite Guinand–Weil dictionary of Groskin provides complementary current context: finite Galerkin coefficient vectors can be transported to explicit band-limited test functions whose finite quadratic values equal exact sums over the nontrivial zeros [5]. Suzuki’s operator formulation gives a continuous spectral treatment of the localized Weil form [7], while recent numerical work demonstrates exponential residual amplification under the synthetic insertion of an off-critical zero [6]. These developments motivate the quantitative questions in section 5, but none is used to prove the finite linear-algebra theorem.

Claim boundary. Nothing in this paper proves the Riemann hypothesis. The separator is local to a single pair block. In a complete zero-side quadratic form, other contributions can mask that block's negative direction. Proving that the explicit certificate survives the full zero population is a separate problem. Likewise, the generic cone-separation theorem is not by itself a theorem about radar, cybersecurity, control, or any other application domain; such uses require an independent model-to-cone interface.

2. FINITE RANK-ONE GEOMETRY

Let V be a finite-dimensional real inner-product space. For $x \in V$, write

$$x \otimes x : V \rightarrow V, \quad (x \otimes x)z = \langle x, z \rangle x.$$

In an orthonormal coordinate system this operator is represented by the symmetric rank-one matrix $xx^\top$. Throughout the abstract real-linear development, $\langle \cdot, \cdot \rangle$ denotes the real inner product. In the Weil specialization of section 6, a complex response is first decomposed into real and imaginary parts; the relevant zero-side block uses a bilinear transpose outer product rather than a Hermitian conjugate-transpose outer product.

Definition 2.1 (Signed pair block). For $x, y \in V$ and $c \in \mathbb{R}$, define

$$B_{c;x,y} = c(x \otimes x - y \otimes y).$$

When the parameters are clear we write simply B .

Definition 2.2 (Gram determinant). For $x, y \in V$, define

$$\text{GramDet}(x, y) := \|x\|^2 \|y\|^2 - \langle x, y \rangle^2.$$

We abbreviate $\Delta = \text{GramDet}(x, y)$.

By Cauchy–Schwarz, $\Delta \geq 0$, with $\Delta = 0$ if and only if x and y are linearly dependent.

Definition 2.3 (Orthogonal witness). Define

$$w(x, y) := \|x\|^2 y - \langle x, y \rangle x.$$

The vector w is the component of y orthogonal to x , multiplied by $\|x\|^2$ so that no denominator is required.

Lemma 2.4 (Witness contractions). For all $x, y \in V$,

$$\langle x, w(x, y) \rangle = 0, \quad \langle y, w(x, y) \rangle = \text{GramDet}(x, y).$$

Proof. Directly,

$$\langle x, w \rangle = \|x\|^2 \langle x, y \rangle - \langle x, y \rangle \|x\|^2 = 0.$$

Similarly,

$$\langle y, w \rangle = \|x\|^2 \|y\|^2 - \langle x, y \rangle^2 = \text{GramDet}(x, y).$$

□

Theorem 2.5 (Explicit Gram witness). Let $x, y \in V$ and let $c \in \mathbb{R}$. For $B = B_{c;x,y}$ and $w = w(x, y)$,

$$(2.1) \quad \langle Bw, w \rangle = -c \text{GramDet}(x, y)^2.$$

In particular, if $c > 0$ and x, y are linearly independent, then

$$\langle Bw, w \rangle < 0.$$

Proof. From the rank-one form of B ,

$$\langle Bw, w \rangle = c \langle x, w \rangle^2 - c \langle y, w \rangle^2.$$

Apply Lemma 2.4. □

Remark 2.6. The argument is deliberately local. It does not diagonalize B and it does not solve an optimization problem. It constructs a negative direction directly from the two response directions x and y .

3. NORMALIZED MARGIN AND EXACT INERTIA

The raw identity (2.1) becomes more informative after separating scale from geometry. Assume $x \neq 0$ and define the orthogonal component

$$(3.1) \quad y_{\perp} := y - \frac{\langle x, y \rangle}{\|x\|^2} x.$$

Then

$$(3.2) \quad w = \|x\|^2 y_{\perp}, \quad \text{GramDet}(x, y) = \|x\|^2 \|y_{\perp}\|^2.$$

Consequently,

$$(3.3) \quad \|w\|^2 = \|x\|^2 \text{GramDet}(x, y).$$

Corollary 3.1 (Normalized margin). *Assume $c > 0$ and $\text{GramDet}(x, y) > 0$. With*

$$\hat{w} := \frac{w}{\|w\|},$$

one has

$$(3.4) \quad \langle B\hat{w}, \hat{w} \rangle = -c \frac{\text{GramDet}(x, y)}{\|x\|^2} = -c \|y_{\perp}\|^2 < 0.$$

Proof. Divide (2.1) by (3.3). □

The normalized form makes clear that the effective detection strength is governed by the squared length of the component of y transverse to x .

Proposition 3.2 (Nonzero spectrum of the pair block). *Assume $c > 0$ and let*

$$a = \|x\|^2, \quad b = \|y\|^2, \quad s = \langle x, y \rangle.$$

If x and y are linearly independent, the two nonzero eigenvalues of B are

$$(3.5) \quad \lambda_{\pm} = \frac{c}{2} \left((a - b) \pm \sqrt{(a + b)^2 - 4s^2} \right).$$

Their product is

$$(3.6) \quad \lambda_+ \lambda_- = -c^2 \text{GramDet}(x, y).$$

Proof. The range of B is contained in $\text{span}\{x, y\}$. In an orthonormal basis adapted to x and $y_{\perp}$, the restriction has trace $c(a - b)$ and determinant $-c^2 \text{GramDet}(x, y)$. The quadratic formula gives (3.5). □

Corollary 3.3 (Exact isolated-pair inertia). *If $c > 0$ and x, y are linearly independent, then*

$$\text{Inertia}(B) = (1, 1, \dim V - 2),$$

where the entries denote the numbers of positive, negative, and zero eigenvalues.

Proof. By (3.6), the two nonzero eigenvalues have opposite signs. Linear independence implies $\text{rank } B = 2$. □

Remark 3.4 (Relation to negative-index results). Corollary 3.3 concerns the isolated signed pair block, not a complete finite Weil matrix. Bombieri's theorem is substantially broader at the finite spectral level: it controls the inertia of the full finite matrix associated with a finite multiset carrying the zeta symmetries [1]. The role of theorem 2.5 is to localize one unit of negative geometry by an explicit vector and exact value.

4. EXPLICIT SEPARATION FROM THE POSITIVE-SEMIDEFINITE CONE

Let $\text{Sym}(V)$ denote the real self-adjoint operators on V and let

$$\mathbb{S}_+(V) := \{P \in \text{Sym}(V) : \langle Pz, z \rangle \geq 0 \text{ for all } z \in V\}$$

be the positive-semidefinite cone. In finite dimensions,

$$(4.1) \quad \mathbb{S}_+(V) = \left\{ \sum_{j=1}^r c_j (v_j \otimes v_j) : r < \infty, c_j \geq 0 \right\}.$$

For $w \in V$, define the linear functional

$$(4.2) \quad \ell_w : \text{Sym}(V) \rightarrow \mathbb{R}, \quad \ell_w(A) := \langle Aw, w \rangle.$$

Lemma 4.1 (Nonnegativity on the on-model cone). *For every $P \in \mathbb{S}_+(V)$ and every $w \in V$,*

$$\ell_w(P) \geq 0.$$

Theorem 4.2 (PSD-cone separator). *Assume $c > 0$ and $\text{GramDet}(x, y) > 0$. Let $B = B_{c;x,y}$ and $w = w(x, y)$. Then*

$$\ell_w(P) \geq 0 \quad \text{for every } P \in \mathbb{S}_+(V),$$

while

$$\ell_w(B) = -c \text{GramDet}(x, y)^2 < 0.$$

Consequently,

$$(4.3) \quad B \notin \mathbb{S}_+(V).$$

More strongly, ℓ_w is an explicit separating linear functional between B and the whole cone (4.1).

Proof. The first assertion is Lemma 4.1; the second is Theorem 2.5. If B belonged to $\mathbb{S}_+(V)$ the two inequalities would contradict one another. $\square$

Remark 4.3. The representation statement repackages the same indefiniteness as a model-class obstruction: no change of nonnegative coefficients and no enlargement of the collection of real rank-one atoms can reproduce B inside the allowed cone. The explicit functional ℓ_w provides the certificate.

5. ROBUSTNESS AND MASKING

The explicit witness produces a quantitative error budget. Throughout this section, assume $c > 0$ and $\text{GramDet}(x, y) > 0$, and write

$$(5.1) \quad m := -\langle B\hat{w}, \hat{w} \rangle = c \frac{\text{GramDet}(x, y)}{\|x\|^2} > 0.$$

Corollary 5.1 (Operator-norm robustness). *Let $E \in \text{Sym}(V)$. If*

$$(5.2) \quad \|E\|_{\text{op}} < m,$$

then

$$\langle (B + E)\hat{w}, \hat{w} \rangle < 0.$$

The same conclusion follows from the stronger but often convenient condition $\|E\|_F < m$.

Proof. For a unit vector $\hat{w}$,

$$|\langle E\hat{w}, \hat{w} \rangle| \leq \|E\|_{\text{op}}.$$

Therefore

$$\langle (B + E)\hat{w}, \hat{w} \rangle \leq -m + \|E\|_{\text{op}} < 0.$$

Finally $\|E\|_{\text{op}} \leq \|E\|_F$. $\square$

The same calculation gives an exact one-direction masking criterion when the background is positive.

Corollary 5.2 (Exact witness masking criterion). *Let $P \in \mathbb{S}_+(V)$ and $M = P + B$. Then*

$$(5.3) \quad \langle Mw, w \rangle < 0 \iff \langle Pw, w \rangle < c \text{GramDet}(x, y)^2.$$

Equivalently,

$$(5.4) \quad \langle P\hat{w}, \hat{w} \rangle < m \iff \langle M\hat{w}, \hat{w} \rangle < 0.$$

Proof. Use $\langle Bw, w \rangle = -c \text{GramDet}(x, y)^2$ and add $\langle Pw, w \rangle$. $\square$

Thus the masking problem is localized to a scalar inequality along an explicitly known direction. A coarse sufficient condition is

$$(5.5) \quad \|P\|_{\text{op}} < m.$$

The criterion (5.4) is sharper because it uses only the background energy seen by the witness.

Proposition 5.3 (Abstract amplification-to-separation principle). *Let $t \mapsto B_t$ be a family of transverse signed pair blocks with normalized witnesses $\hat{w}_t$ and margins*

$$m(t) := -\langle B_t \hat{w}_t, \hat{w}_t \rangle > 0.$$

Let $P_t \in \mathbb{S}_+(V_t)$. If

$$\langle P_t \hat{w}_t, \hat{w}_t \rangle = o(m(t)) \quad (t \rightarrow \infty),$$

then for all sufficiently large t ,

$$\langle (P_t + B_t) \hat{w}_t, \hat{w}_t \rangle < 0.$$

A sufficient condition is $\|P_t\|_{\text{op}} = o(m(t))$.

Proof. Divide (5.4) by $m(t)$ and use the assumed $o(1)$ ratio. $\square$

Remark 5.4 (What remains hard in the zeta problem). The proposition is an abstract dominance lemma, not an RH result. In the zeta setting, one must justify a probe family, identify the target pair, and control the full population of other off-critical and critical-line contributions without assuming a statement equivalent to RH. Recent numerical work observes $e^{2a\delta}$ amplification after inserting a synthetic zero with real displacement δ [6]; turning such amplification into the hypothesis of Proposition 5.3 for the exact full zero population is a separate analytic problem.

6. SPECIALIZATION TO FINITE WEIL PAIR BLOCKS

We now record the interface through which the abstract theorem enters the finite Weil setting used in the accompanying formalization. The purpose is not to reconstruct the Guinand–Weil explicit formula from scratch; see [8, 1, 2] and the recent finite dictionary [5] for that background.

Let a finite bank of real probes produce, at a spectral point, a complex response vector

$$u = x + iy \in \mathbb{C}^d, \quad x, y \in \mathbb{R}^d.$$

For the reflection-symmetric partner, the response is conjugated. The relevant zero-side construction is bilinear in the response vectors: it uses transpose $(\cdot)^\top$, not Hermitian conjugate

transpose $(\cdot)^*$. Consequently, at the block level, summing the response outer product with its conjugate gives

$$\begin{aligned} uu^\top + \bar{u}\bar{u}^\top &= (x + iy)(x + iy)^\top + (x - iy)(x - iy)^\top \\ (6.1) \qquad \qquad &= 2(xx^\top - yy^\top). \end{aligned}$$

With multiplicity $m > 0$, set $c = 2m$. The real pair block is therefore precisely

$$(6.2) \qquad \qquad B_\rho = c(xx^\top - yy^\top).$$

This is the pair-block identity used in the Lean source described in section 9.

On the critical line, the corresponding response vector is real in this model, so the imaginary component vanishes and the block reduces to a nonnegative real rank-one matrix. Finite sums of such contributions lie in the cone (4.1).

Definition 6.1 (Finite visibility). A pair is *visible to the probe bank* if its real and imaginary response vectors are transverse:

$$\text{GramDet}(x, y) > 0.$$

Theorem 6.2 (Finite separation of a visible off-critical pair). *Suppose an off-critical reflection pair has finite response $u = x + iy$ satisfying Definition 6.1, and let B_ρ be its block (6.2). Then the explicit vector*

$$w = \|x\|^2 y - \langle x, y \rangle x$$

satisfies

$$w^\top B_\rho w = -c \text{GramDet}(x, y)^2 < 0.$$

For every finite on-line configuration

$$P = \sum_j c_j v_j v_j^\top, \quad c_j \geq 0, \quad v_j \in \mathbb{R}^d,$$

one has

$$w^\top P w \geq 0.$$

Consequently B_ρ cannot be represented by any such on-line configuration.

Proof. Apply theorems 2.5 and 4.2 to (6.2). □

Remark 6.3 (Locality). Theorem 6.2 is a block-level result. If the complete finite zero-side matrix is written schematically as

$$G = B_\rho + R,$$

then $w^\top G w < 0$ requires an additional bound on $w^\top R w$. The theorem does not assert such a bound. This is the masking problem isolated in Corollary 5.2.

7. RELATION TO BOMBIERI'S FINITE INERTIA THEOREM

Bombieri's 2000 analysis is the closest classical finite negative-index result. For a finite multiset with the symmetries of the transformed zeta-zero set, Bombieri constructs finite matrices $H(\Gamma; t)$ and proves that the number of negative eigenvalues equals the number of distinct complex-conjugate pairs [1, Theorem 8]. He obtains the result by a deformation argument: the spectral points are moved while zero is excluded as an eigenvalue, and an added complex symmetry package is analyzed in a regime where the new eigenvalues separate asymptotically into positive and negative branches. He then passes from finite systems toward the zeta setting under additional hypotheses.

The present theorem does not improve Bombieri's global inertia count. Instead it provides a different type of information about one pair once its finite response vector has been isolated. The distinction is summarized in table 1.

TABLE 1. Bombieri’s finite negative-index result versus the present pair-local certificate.

Feature	Bombieri 2000	Present result
Finite negativity from complex/off-critical pairs	Yes	Yes
Negative-inertia count for $H(\Gamma; t)$	Yes	Not the objective
Method	Deformation and eigenvalue analysis	Direct Gram geometry
Pair-local signed block	Not the central formulation	$c(xx^\top - yy^\top)$
Closed-form negative witness	Not used in the audited argument	Explicit w
Exact pair-local negative value	Not used in the audited argument	$-c \text{GramDet}(x, y)^2$
Separation from PSD rank-one cone	Not used in the audited argument	Explicit
Perturbation/masking margin	Not used in the audited argument	Explicit
Lean formalization	No	Core theorem suite

The appropriate interpretation is therefore a *constructive localization* of a finite negativity phenomenon whose existence and inertia count are classical.

8. RELATION TO RECENT FINITE AND OPERATOR FORMULATIONS

8.1. Finite Guinand–Weil dictionaries. Groskin constructs an exact finite transport from real even Galerkin coefficient vectors to band-limited Guinand–Weil test functions and proves that every corresponding cutoff-free finite quadratic value equals the exact sum of that test over the nontrivial zeta zeros [5]. That result gives a calibrated finite instrument and carefully distinguishes exact cutoff-free signs from finite-cutoff tail artifacts. The present pair separator is complementary: it concerns the geometry of an individual hypothetical off-critical contribution after a finite probe response has been specified.

8.2. Localized Weil operators and screw functions. Suzuki gives an explicit screw-function realization of localized Weil quadratic forms and a self-adjoint operator framework that unifies earlier variational and spectral results [7]. Failure of RH corresponds to negativity of a localized lowest spectral value for some scale, while additional results concern the reality of zeros of characteristic functions associated with finite or localized operators. This is an operator-level manifestation of Weil positivity rather than a closed pair-local certificate.

8.3. Amplification of a synthetic off-line mode. Kim et al. numerically realize Suzuki’s operator and study a residual in which a synthetically injected zero at real displacement $\delta > 0$ from the critical line produces exponential growth with scale $e^{2a\delta}$ [6]. Their work explicitly treats this as a numerical/operator realization of classical Weil-criterion behavior, not as a proof of RH. The quantitative pairing with Corollary 5.2 and Proposition 5.3 is natural: an explicit negative margin becomes more useful if a probe parameter can amplify the target mode while the masking background remains controlled. Establishing such a dominance theorem for the genuine complete zero population remains open.

8.4. Finite real-zero transfer. Connes and van Suijlekom prove real-zero localization results for Fourier transforms of special eigenfunctions associated with lower-bounded quadratic forms, proceeding through structured finite matrices and a Hurwitz limit argument [4]. Their theorem addresses a different finite-to-infinite mechanism. It reinforces a general methodological point: finite structure is useful only when paired with an exact theorem controlling how that structure survives a limiting or spectral passage.

9. FORMAL VERIFICATION IN LEAN 4

The core pair-block theorem suite is formalized in the public repository

`HMarcusWH/zeta-23-lean`

at the source snapshot

`a47095149377aeec01a0335945fcdcd311f0444.`

The exact file is `Zeta23/ExceptionalZero/ProbeGramNegativity.lean`. Its blob SHA at this snapshot is

`ecc4b574a657c3e90331440a18cbfdefe32faa70.`

A pinned browser view is available at

`github.com/HMarcusWH/zeta-23-lean/.../ProbeGramNegativity.lean.`

The repository snapshot pins Lean to `v4.33.0-rc2` and Mathlib to commit

`51e6992efd06126df61a496bebf8f49482a4e129.`

TABLE 2. Paper objects and corresponding Lean declarations.

Paper object/result	Lean identifier
$B = c(xx^\top - yy^\top)$	<code>pairBlock</code>
$\text{GramDet}(x, y)$	<code>gramDet</code>
$w = \ x\ ^2 y - \langle x, y \rangle x$	<code>orthoWitness</code>
$\langle x, w \rangle = 0$	<code>dotProduct_orthoWitness_left</code>
$\langle y, w \rangle = \text{GramDet}(x, y)$	<code>dotProduct_orthoWitness_right</code>
Exact identity $\langle Bw, w \rangle = -c\Delta^2$	<code>dotProduct_pairBlock_orthoWitness</code>
Strict negativity	<code>dotProduct_pairBlock_orthoWitness_neg</code>
Nonnegativity for finite on-line rank-one sums	<code>dotProduct_nonneg_of_onLineCombination</code>
Representation obstruction	<code>not_onLineCombination_of_pairBlock</code>

The formalization represents real vectors as complex vectors fixed by conjugation, allowing the same matrix infrastructure to interface directly with the zero-side model. The comments in the source explicitly record the claim boundary: the theorem separates one off-line pair block from any on-line nonnegative rank-one configuration, but does not assert that the pairwise negative direction survives the complete on-line bulk.

Formal-status boundary. The declarations listed in table 2 are present in the cited Lean snapshot. The normalized-margin formula, exact inertia corollary, and perturbation/masking statements in sections 3 and 5 are direct paper corollaries of the formalized identity, but they are not claimed here to be separate Lean declarations in that snapshot.

10. PRIORITY BOUNDARY AND NOVELTY STATEMENT

The broad mathematical phenomenon is classical: Weil positivity is equivalent to RH, and Bombieri’s finite theory already proves negative inertia associated with complex pairs [8, 1]. The generic linear algebra used in theorem 2.5 is elementary once the signed block (1.1) is written down. Accordingly, no claim is made that theorem 2.5 constitutes a fundamentally new theorem of linear algebra.

The narrower claim is about the constructive finite formulation in the Weil pair-block setting. In a targeted source-level audit of the closest materials available to the author—including Bombieri’s finite-inertia argument, the recent finite Guinand–Weil dictionary, Suzuki’s screw-function formulation, recent numerical operator work, and the contemporary Connes program—we did not identify a prior formulation combining all of the following:

- (a) the explicit pair-local block $c(xx^\top - yy^\top)$ as the separator object;
- (b) the closed witness $w = \|x\|^2 y - \langle x, y \rangle x$;
- (c) the exact value $-c \text{GramDet}(x, y)^2$;
- (d) the explicit representation obstruction against the entire nonnegative real rank-one cone.

We therefore describe the result conservatively as an *apparently new constructive local refinement* of classical finite Weil negativity. This is a “to our knowledge” statement, not a claim of exhaustive universal priority. In particular, the result should be read against Bombieri’s variational work, Yoshida’s Hermitian-form framework, and adjacent finite-negative-index literature [2, 9]. No exhaustive priority claim is made beyond the sources explicitly audited.

11. OPEN PROBLEMS

The finite certificate suggests several precise next questions.

- P1. Minimal probe dimension.** Determine the smallest finite probe bank for which a visible off-critical pair can be forced to have $\text{GramDet}(x, y) > 0$. Two probes may suffice generically, but exceptional phase configurations require exact treatment.
- P2. Constructive transversality.** For translated or otherwise parametrized probes, derive explicit conditions on the probe parameters that guarantee $\text{GramDet}(x, y) > 0$ and quantify the resulting margin m .
- P3. Full-zero masking.** Given one target pair, control the remaining zero population in the explicit direction $\hat{w}$. The exact target is the inequality in (5.4), not merely a global operator-norm estimate.
- P4. Amplification crossover.** Combine an off-line amplification mechanism with the explicit margin and prove a rigorous form of

$$\text{target amplification} \implies \text{eventual unmasked finite negativity,}$$

without assuming a bound equivalent to RH.

- P5. General L -functions.** Identify which explicit-formula settings preserve the same reflection-pair signed-rank-one geometry and admit the same separator.
- P6. Formal strengthening.** Add Lean declarations for the normalized margin, exact inertia, perturbation stability, and masking criterion, so that every theorem-level quantitative statement in this paper is machine checked.

A useful discipline for any proposed RH application is to distinguish an actual weakening of the missing theorem from an equivalent reformulation. The present paper deliberately stops at the finite structural statement and the exact masking inequality.

12. CONCLUSION

Bombieri’s finite theory shows that complex zero pairs generate negative inertia. The result developed here localizes that phenomenon in a finite probe-response block. For

$$B = c(xx^\top - yy^\top),$$

the vector

$$w = \|x\|^2 y - \langle x, y \rangle x$$

is an explicit negative certificate with exact value

$$w^\top B w = -c \text{GramDet}(x, y)^2.$$

The same vector defines a separating functional from the full positive-semidefinite rank-one cone, yields a normalized robustness margin, and reduces pairwise masking to a scalar inequality in one known direction. The core witness and separation theorems are formally verified in Lean 4.

The theorem is not an RH proof and does not replace the global finite-inertia results already in the literature. Its value is constructive: it turns a visible pair’s negative geometry into a calculable finite certificate. Whether a suitable probe family can force this certificate to survive the complete zeta zero population remains open.

APPENDIX A. COORDINATE CALCULATION

For readers preferring matrices, let $x, y, w \in \mathbb{R}^d$ and

$$B = c(xx^\top - yy^\top).$$

Then for any $z \in \mathbb{R}^d$,

$$(A.1) \quad z^\top B z = c(x^\top z)^2 - c(y^\top z)^2.$$

Set

$$w = (x^\top x)y - (x^\top y)x.$$

A direct calculation gives

$$x^\top w = 0,$$

and

$$y^\top w = (x^\top x)(y^\top y) - (x^\top y)^2 = \text{GramDet}(x, y).$$

Substitution into (A.1) yields

$$w^\top B w = -c \text{GramDet}(x, y)^2.$$

This is the entire algebraic core of the separator.

APPENDIX B. FORMAL THEOREM CORRESPONDENCE

The Lean source uses complex-valued finite vectors with hypotheses asserting that x and y are fixed by conjugation. In that representation the key declaration has the schematic form

```
theorem dotProduct_pairBlock_orthoWitness ... :
  star (orthoWitness x y) dot
    (pairBlock c x y *v orthoWitness x y)
  = -(c : C) * (gramDet x y * gramDet x y)
```

followed by a strict-negativity theorem under $c > 0$ and nonzero Gram determinant, a positive-semidefinite theorem for nonnegative sums of real rank-one blocks, and the representation-obstruction theorem. The complete compiler-readable source is linked in section 9.

APPENDIX C. REPRODUCIBILITY RECORD

Repository: <https://github.com/HMarcusWH/zeta-23-lean>
Source snapshot: a47095149377aeec01a0335945fcdcde311f0444
Formal source: Zeta23/ExceptionalZero/ProbeGramNegativity.lean
Source blob: ecc4b574a657c3e90331440a18cbfdefe32faa70
Lean: leanprover/lean4:v4.33.0-rc2
Mathlib: 51e6992efd06126df61a496bebf8f49482a4e129

The commit hash and source blob above identify immutable Git objects. Any archival release should preserve the same snapshot together with the build instructions, independently of the moving default branch.

Data and code availability. No empirical dataset is required for the theorem. The Lean 4 source for the formally verified core is publicly available in the repository and pinned source snapshot listed in section C.

Competing interests. The author declares no competing interests.

Funding. The author reports no external funding for this work.

Use of AI tools. Language-model tools were used for drafting, code assistance, proof organization, and literature-search support. The author takes responsibility for the mathematical statements, proofs, references, claim boundaries, and released formal verification. AI systems are not authors.

REFERENCES

- [1] Enrico Bombieri. “Remarks on Weil’s quadratic functional in the theory of prime numbers. I”. In: *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* 11.3 (2000), pp. 183–233.
- [2] Enrico Bombieri. “A variational approach to the explicit formula”. In: *Communications on Pure and Applied Mathematics* 56.8 (2003), pp. 1151–1164. DOI: 10.1002/cpa.10089.
- [3] Alain Connes and Caterina Consani. “Spectral triples and ζ -cycles”. In: *L’Enseignement Mathématique* 69.1–2 (2023), pp. 93–148. DOI: 10.4171/LEM/1049.
- [4] Alain Connes and Walter D. van Suijlekom. “Quadratic Forms, Real Zeros and Echoes of the Spectral Action”. In: *Communications in Mathematical Physics* 406 (2025). Article 312. DOI: 10.1007/s00220-025-05493-1. arXiv: 2511.23257.
- [5] Akiva Groskin. *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*. 2026. arXiv: 2607.02828 [math.NT].
- [6] Taebong Kim, Youngsik Hong, Minsik Kim, Sunyoung Choi, Jaewon Jang, and Minseo Kim. *A Numerical Realization of Suzuki’s Weil-Quadratic-Form Operator: The Archimedean Spectral Law, its Universality, and an Operator Form of Weil’s Positivity Criterion*. 2026. arXiv: 2607.24830.
- [7] Masatoshi Suzuki. *Weil’s quadratic form via the screw function*. 2026. arXiv: 2606.09096 [math.NT].
- [8] André Weil. “Sur les “formules explicites” de la théorie des nombres premiers”. In: *Comm. Sém. Math. Univ. Lund, Tome Supplémentaire*. 1952, pp. 252–265.
- [9] Hiroyuki Yoshida. “On Hermitian forms attached to zeta functions”. In: *Zeta Functions in Geometry (Tokyo, 1990)*. Vol. 21. Advanced Studies in Pure Mathematics. Tokyo: Kinokuniya, 1992, pp. 281–325.

Email address: marcus@hmwh.se